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Model for the injection of charge through the contacts of organic transistors

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A compact model has been employed in organic thin film transistors (OTFTs) to study the electrical characteristics of the contacts, which are formed between the organic layer and source/drain electrodes of the OTFT. The model shows the importance of interrelating different physical phenomena: charge injection, redox reactions at the interface, and charge drift in the organic semiconductor. The model reproduces and explains several features that have been reported for current-voltage curves, I_D-V_C, at the contacts of OTFTs. The I_D-V_C curves are extracted from the experimental output characteristics by two techniques. One technique uses a set of transistors with different channel lengths and a simultaneous extraction of the I_D-V_C curve and the mobility of carriers in the channel of the transistor. When a set of transistors with different channel lengths is not available, we propose an iterative method for the simultaneous extraction of the I_D-V_C curve and the mobility by changing the gate bias voltages. © *2009 American Institute of Physics*.
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I. INTRODUCTION

Organic or polymeric (hereafter, the term organic is used for both) thin film transistors (OTFTs) attracted the attention of many researchers during the last decade. A major application for these transistors is in low-end, high-volume microelectronics such as identification tags where mechanical flexibility and low cost are necessary. These devices are also suitable for large-area flexible applications[1] and present low-cost, low temperature processing such as dipping or spin-coating technologies.[2] OTFTs are being considered for the drive electronics in liquid crystal display[3] and organic light-emitting diodes, and might also be applied to smart cards, gas sensors, and other fields.

A key element in OTFTs research has been the organic semiconductor material itself. However, when the lateral potential distribution in the channels of OTFTs was mapped in Ref. 4, it was found that a substantial part of the externally applied drain-source voltage V_{DS} drops across the contact between the organic material and metal electrodes for source and drain, in particular, between the source and the organic material. Thus, in addition to the limited drift in the organic film, there is one more limit for the charge flow in OTFTs owing to contact effects.

As a conclusion of potentiometry experiments, several authors[5–7] consider that the current finds two different resistive regions in the path from the source to the drain, consequently causing a voltage drop at the contacts and a voltage drop along the intrinsic channel. The "parasitic" voltage drop at the contacts leads to a lowering of the effective bias voltage applied to the intrinsic drain-source channel of the transistor and, consequently, reduces the values for current and mobility.[8,9] The relative importance of contact resistance increases with increasing charge carrier mobility in the organic layer or decreasing the channel length L ($L<1$ μm).[7,8,10] The contact resistance may, therefore, become comparable to or even greater than the channel resistance in short channel devices or in devices with high mobility organic materials in the channel. Thus, it is very important to consider the contact effects when extracting the parameters of the transistors (i.e., mobility and threshold voltage) from the output characteristics of the device.[6,8]

Finding a model for OTFTs is not an easy task. The contact effects and other limiting mechanisms for the charge transport cannot be explained by the silicon metal-oxide-semiconductor field-effect transistor (MOSFET) model.[8] Uncommon output characteristics are associated with gate-voltage dependent contact resistances and mobilities[4,5,7,11–13] or with the topology of the transistor. Several studies[4,8,14] have shown that OTFTs with bottom contact (BC) design demonstrate inferior performance, compared to those with the top contact (TC) design. As a consequence of the problems cited above, the contact effects should be characterized.

Physically based models are required to interpret the dependence of drain current on contact voltage.[5] Models devoted to reproducing current-voltage curves at the contacts of OTFTs show some controversies, mainly because two theories clearly compete between each other: space-charge limited conduction (SCLC) and injection-limited conduction. Researchers who favor one theory neglect the other by demonstrating problems to reproduce certain dependences of the

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current with the applied voltage, the temperature, dimensions of the device, or different kinds of contacts. The limitations of the reported approaches have their origin in the fact that they focus only on a particular phenomenon and therefore explain only certain I_D-V_C dependences. To address these limitations and provide a coherent description of the phenomena present in the contacts of OTFTs, we propose a model that combines the key mechanisms and consequently can reproduce a wider range of experimental data. Our model assumes that the voltage drop at the contact V_C is the superposition of different components. Each component is related to a different mechanism in the contact: injection, redox reactions at the interface, and drift in the adjacent organic material. The relation of these phenomena with the voltage drop at the contacts is the main goal of this study.

This paper is organized as follows: In Sec. II, we review the different models found in the literature that report on the contact effects. The strengths and weaknesses of each model are discussed. Then, we propose a model that combines all physical phenomena. In Sec. III, we focus on the characterization techniques related to the proposed model by reviewing and discussing the different characterization approaches. This allows extracting of the parameters of the transistor and the contact voltage drop from the output characteristics. Finally, a validation of our model by the characterization of several OTFTs is presented in Sec. IV.

II. MODELING OF THE CONTACT EFFECTS

The objective, when modeling the contacts of OTFTs, is the reproduction of the I_D-V_C curves and their dependences on bias voltages, temperature, and material parameters.[4,15–18] The two main theories used for the explanation of contact effects in OTFTs are the injection-limited conduction and SCLC theories. A review of models based on these theories is presented below.

A. Injection limited conduction

In Ref. 19, a study of the transport of charge in a variable thickness layer of tris(8-hydroxyquinoline) aluminum (Alq3) sandwiched between two metal electrodes was described. In order to explain their experimental curves, they proposed that the transport is injection-limited and that the injection is limited by charge hopping where interfacial dipoles play a crucial role because they may alter the effective injection barrier. They eliminated other models, as explained below. At a constant electric field, the bulk-limited models predict a dependence of current density J on the thickness of the organic layer d according to $J \propto 1/d^x (x \geq 1)$. This dependence was not observed in Ref. 19 and therefore they eliminated the bulk-limited model. Another reason for ruling out bulk-limited models is that they do not predict the different characteristics for different cathodes that were observed in Ref. 19. In Ref. 19 the authors also eliminated injection models that depend only on the energy barrier at the interface because the cathode dependence of current-voltage characteristics at low temperatures is substantially reduced. They proposed that the injection is limited by charge hopping where interfacial dipoles play a crucial role as they may alter the effective injection barrier. However, they left an open question on the origin of the interfacial dipoles that lead to a lowering of the interface energy barrier.

Another proposal of injection limited model is in Refs. 17 and 20. The model in Refs. 17 and 20 is based on thermionic emission with diffusion-limited injection currents and was used in Ref. 10 to interpret contact resistances in BC poly(3-hexylthiophene) (P3HT) OTFTs with Au as source and drain contact material. This model was dismissed in Ref. 21 because it could not reproduce the dependence with the temperature for OTFTs having high injection barriers between the organic layer and source/drain electrodes, such as the contacts in OTFTs formed between Cr or Cu with P3HT.[21] In order to reproduce the dependence with the temperature for OTFTs with high injection barriers, a more sophisticated model of hopping injection into a disordered density of localized states was used later.[21]

B. Space charge limited conduction

In TFTs with relatively small contact effects, the contact resistance is found to be dominated by the mobility of the polymer.[4] This suggests that bulk drift dominates in the contact. On the other hand, small contact effects produce a voltage drop in the contact that depends almost linearly with the current flowing through the contact, i.e., nonlinear effects in the output characteristics of the transistor are nearly negligible (Fig. 2 in Ref. 21). This means that bulk drift models like the classical Mott–Gurney law,[22,23] where the dependence of the current density with the applied voltage is nonlinear, could not reproduce experimental characteristics of the contacts of OTFTs.[7,21]

C. Unified model for contacts in OTFTs

None of the aforementioned models can individually explain certain experimental data. The linearity of the current-voltage curves at the contacts I_D-V_C in Refs. 7 and 21 can be explained neither by the drift model in Refs. 22 and 23 nor by the injection limited models.[17,19,20] The drift and the injection limited models have been used for contacts with nonlinear responses, not for devices showing linear I_D-V_C curves. If the contact region thickness increases over 100 nm, transport must eventually become bulk limited[19] and the injection models have to be combined with drift models. Consequently, some combination of the models should be studied. In addition, it is also necessary to understand the chemistry at the metal/organic interface and the physical origins of the interface dipoles that lead to a lowering of the interface energy barrier.[19] Therefore, we find as appropriate a unified model for contacts in OTFTs, which takes into account all the physical phenomena present in the contacts of OTFTs, i.e., injection, redox reactions at the interface, and drift in the bulk of the contact.

The unified model was developed previously to explain the contact effects in organic diodes.[24] Considering that the same principles for the contact effects in organic diodes are valid for other organic devices,[5] the unified model is now applied to the depleted region located between the source and the accumulation layer in the organic material of OTFTs.

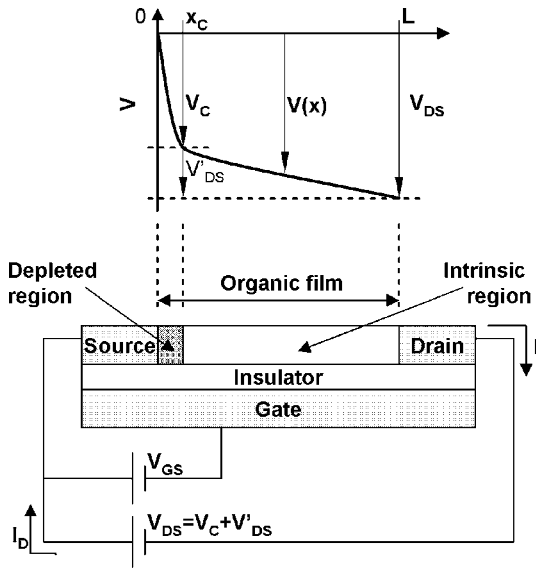

FIG. 1. Separation of the channel of a BC organic transistor in two regions: the contact region $[0, x_C]$ modeled by a voltage drop V_C and the intrinsic channel $[x_C, L]$.

This region includes the source contact and a low conductivity region[25] close to it (Fig. 1). There is not only electrical evidence of the depleted region [as was demonstrated in (Ref. 4) by means of scanning potentiometry], but also structural evidence of this region as was found in Ref. 26. They detected a transition region from large grain pentacene in the bulk of the channel of Au-pentacene OTFTs to a microcrystalline form at the interface between the metal electrode and the organic film. The voltage drop at the drain contact is much less important than the voltage drop at source as supported by scanning potentiometry[4] and will therefore be ignored.

Our model assumes that three phenomena take place in charge transport through the depleted region of the OTFT:

- Charge carriers pass a potential barrier at the contacts.
- Ions are created at the electrodes by a redox reaction in the organic material. These ions form charged dipoles that modulate the injection barrier and are capable of accepting other hopping electrons.
- Drift of charge carriers in the bulk of the contact region.

Thus, to maintain a current I_D, the contact voltage V_C is shared by voltages required for carrier injection through the barrier $V_{\text{injection}}$, for ion formation (redox reactions) V_{redox} and for charge drift V_{drift}. The relationship between the contact voltage V_C and the current through the structure is described by the equation,

$$V_C = V_{\text{injection}}(I_D) + V_{\text{redox}}(I_D) + V_{\text{drift}}(I_D). \tag{1}$$

The different components are given in Eqs. (2)–(4) below.

The voltage for the redox process V_{redox}, according to Nernst equation,[24] is

$$V_{\text{redox}} = \Phi + V_t(T)\ln(\rho_{dp}/\rho_m), \tag{2}$$

where ρ_{dp} is the value of the ion density forming the dipoles at the interface between the metal and the organic material, Φ is a constant which depends on the materials involved in a given redox reaction, ρ_m is the molecular density of the organic material, and $V_t(T) \equiv kT/e$ is the thermal voltage, where T is the absolute temperature, $k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant, and $e = 1.6 \times 10^{-19}$ C is the magnitude of the electron charge.

The voltage required for the transport across the contact region is calculated assuming that the charge transport is due to the drift of positive charges (we focus on p-type OTFTs) in a space-charge region with charge density at the interface $p(x=0)=p(0)$. By this boundary condition at the contact interface, it is shown in Ref. 24 that the voltage drop associated with the charge drift is

$$V_{\text{drift}} = \frac{2}{3}\left\{\frac{2J}{\varepsilon\mu}\right\}^{1/2}\left\{[x_c + x_p]^{\frac{3}{2}} - [x_p]^{\frac{3}{2}}\right\},$$

$$x_p \equiv \frac{J\varepsilon}{2e^2\mu[p(0)]^2}, \tag{3}$$

where $J = I_D/S$ is the current density, S is the cross section of the metal/organic contact, $S = t_o W$, t_o is the thickness of the organic layer, W is the channel width of the OTFT, x_c is the contact length equal to the length of the depleted region as shown in Fig. 1, ε is the permittivity of the organic material, and μ is the mobility of the carriers. The parameter x_p is a characteristic length (from contact interface toward organic film), in which the charge density $p(0)$ at the metal/organic interface decays to $p(0)/\sqrt{2}$. In the first-order of approximation, $p(0) \equiv \rho_{dp}$, where ρ_{dp} is the ion density in Eq. (2) for the redox voltage V_{redox}.

Equation (3) allows for the incorporation of a variety of mobility models for the charge carriers, e.g., a variable range hopping model[16] or band conduction model.[27] The voltage required for the injection $V_{\text{injection}}$ could be evaluated using the injection models[17,19,20,24,28] discussed in Sec. II A. In general, charge injection through metal-organic junction can be due to thermionic emission and tunneling.[24] However, in OTFTs, the electric field at the contact is not very high, therefore, we consider that the injection is only due to thermionic emission, given by[24]

$$I_D = A^* T^2 S \exp\left(\frac{-\varphi_M}{kT}\right)\exp\left(\frac{V_{\text{redox}}}{V_t(T)}\right)\left[\exp\left(\frac{V_{\text{injection}}}{\eta V_t(T)}\right) - 1\right]$$
$$= I_S\left[\exp\left(\frac{V_{\text{injection}}}{\eta V_t(T)}\right) - 1\right], \tag{4}$$

where $A^* = 120$ A cm^{-2} K^{-2} is the Richardson constant, φ_M is the height of the barrier between the source metal and the organic layer, I_S is the saturation current, and η is ideality factor of the junction. The barrier height φ_M is calculated as the difference between the metal work function and the organic material's highest occupied molecular orbital (HOMO).[1,6,18,29] V_{redox} is the barrier reduction owing to dipoles at the interface between the source region and the metal.[19] The barrier reduction is accounted by V_{redox}. Larger V_{redox} increases the saturation current I_S by $\exp[V_{\text{redox}}/V_t(T)]$, see again Eq. (4).

III. CHARACTERIZATION TECHNIQUES

An important problem in OTFTs research is the separation of contact effects from the characteristics of the intrinsic transistor. The separation is not a trivial task because the contact effects interfere with other dependences in OTFTs.[8] For example, the mobility and the threshold voltage extracted by commonly used techniques based on the MOSFET model are influenced by the contacts and, as a result, the OTFT data need further processing. In this section, we discuss approaches to extract the current versus voltage curve at the contact I_D-V_C, the threshold voltage V_T, and the charge carrier mobility μ.

It was considered[5,14,30] that the mobility and/or contact resistance are gate voltage dependent. However, different parameters are attributed to be dependent on the gate voltage as discussed below.

For Au-sexithiophene OTFTs, a method assuming a constant contact resistance and a gate voltage dependent mobility was developed to extract the mobility from the transfer characteristics and to estimate values for the threshold voltage and contact resistance.[30,31] For Au-pentacene OTFTs, a model with a gate voltage dependent mobility and highly nonlinear drain and source contact resistances was proposed to simulate the transport characteristics.[14]

On the other hand, in OTFTs based on polyfluorene F8T2, which makes a non-Ohmic contact with Au, the mobility has been assumed constant and the nonlinear current-voltage relation at the source has been considered dependent on the gate voltage.[5] The contact resistance was extracted from the output characteristics of transistors with different channel lengths and the contact current was found to be exponentially related to the contact voltage. It is also shown in Ref. 6 that the method can be used to simultaneously extract the mobility and the I_D-V_C curve. We will show shortly that an alternative method to extract the I_D-V_C curves and mobility can be used when a set of devices with different channel lengths is not available. Overall, the procedures for extraction of contact characteristics can be grouped into two approaches: by variation of channel length or by variation of gate bias voltage. These are now described.

A. Set of transistors with different channel lengths

This procedure is based on the assumption that the channel of an OTFT, such as the p-channel bottom-contact configuration depicted in Fig. 1, can be split into two regions: contact region and intrinsic TFT region. By this assumption, the total source-drain voltage V_{DS} is therefore split into a channel component V_{DS} $(=V_{DS}-V_C)$ and a voltage dropped at the contacts V_C. It is supported by scanning potentiometry[4] that V_C is mainly dropped at the source contact. By using the gradual channel approximation and after integration of the potential along the channel, the current of OTFTs is obtained as[5,6]

$$I_D=\mu W C_{ox}/(L-x_c)[(V_{GS}-V_T)(V_{DS}-V_C) - (V_{DS}^2-V_C^2)/2], \quad (5)$$

where C_{ox} is the gate capacitance per unit area, and the mobility might be unknown, but constant at a given gate voltage.

When identically processed transistors with different channel lengths (but the same channel width) are available, the I_D-V_{DS} curves are measured at a given gate-source voltage V_{GS} for each transistor. The corresponding I_D-V_C is calculated from Eq. (5) for all the different transistors. Assuming that the contact and channel transports are identical in different transistors (independent of L), the correct value of μ would make all the different I_D-V_C curves to coincide when the different samples have the same width.

This technique has been used for different length OTFTs based on several organic materials[6,12,25,31] in BC configuration. It has been shown that both the mobility and the contact resistance are gate voltage dependent.[6,12,25] This characterization technique, however, has two limitations.[7] One is that a series of devices with different channel lengths, but otherwise identical, has to be available, and second, the channel transport properties in different devices might not be exactly the same. In such cases, we propose to use the alternative characterization technique with variable gate biasing, given below.

B. Set of characteristics of a single transistor biased at different gate voltages

In the case when transistors with different channel lengths are not available or the transport depends on L, we propose a method based on the output characteristics at different bias conditions. The details of the procedure to extract the I_D-V_C curves and the mobility are the following.

First, initial estimates for the threshold voltage and the mobility are made by means of the MOSFET model[7,25] by fitting a transfer characteristic in the saturation regime. The initial estimates are the so-called apparent threshold voltage V_{AT} and apparent mobility, μ_A.[7,25]

Second, it is observed that the contact effects only slightly increase the threshold voltage of the OTFT.[5] Therefore, $V_T \approx V_{AT}$ and the initial value for the threshold voltage is assumed also as a final value.

Third, the mobility and the voltage drop at the contact are extracted from the output characteristics by an iterative procedure using Eq. (5). The procedure accounts for the following facts: (i) both the mobility and the contact voltage are gate voltage dependent, and (ii) the apparent mobility μ_A is lower than the mobility μ of the intrinsic OTFT, because μ_A includes contact effects.

To set up the iterative use of Eq. (5), we also define a parameter $\alpha \equiv V_C/V_{DS}=[R_C/(R_{ch}+R_C)]$, which accounts for the ratio between the contact resistance R_C and the channel resistance R_{ch}.[25]

The iterative procedure is performed by using the output characteristic at given gate voltage and then repeated for all gate voltages. The procedure is illustrated in Fig. 2 and begins with the initial values mentioned above, $\alpha_0=0$ and $\mu_0=\mu_{AT}$. A new value μ_1 for mobility is obtained by fitting Eq. (5) to the experimental output characteristics in the linear regime (varying the value for mobility), as shown in Fig. 2(a). Next, a new value α_1 is obtained by fitting Eq. (5) to the

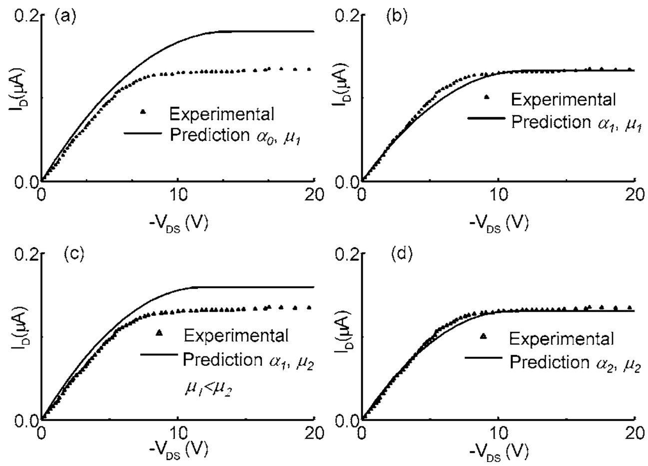

FIG. 2. Iterative procedure to fit output characteristics in an OTFT (symbols) with Eq. (5) (solid lines). For clarity, only a sample curve measured at a given V_{GS} is shown. For other values of $V_{GS}>V_T$ the graphs are similar.

experimental output characteristics in the saturation regime (using μ_1 and varying α), as shown in Fig. 2(b). Then, the above two fitting procedures are repeated alternatively, obtaining the values μ_i and α_i after each loop of number (i) of the iteration. The results from fitting Eq. (5) in linear and saturation regimes in the second loop ($i=2$) are shown in Figs. 2(c) and 2(d), respectively. The above two fitting procedures are repeated until a good agreement with experimental data in linear and linear-saturation transition regions, is reached.

The convergence of the method has been proven good. For example, see again Fig. 2(d). After the second iteration, the model characteristic is almost identical with the measured. Also, as another criterion, it will be illustrated later in Sec. IV B that the above procedure obtains values for mobility that increase with the gate voltage and values for contact resistance that decrease with the gate voltage, as repeatedly observed in experiments, e.g., in Ref. 25.

IV. VALIDATION OF THE MODEL BY CHARACTERIZATION OF DEVICES

In this section, we demonstrate that the unified model interprets consistently I_D-V_C curves of different OTFTs with output characteristics impacted by contact effects. The examples include both characterization methods, set of transistors with different channel lengths at constant gate voltage, and set of characteristics of a single transistor biased at different gate voltages.

We chose experimental data for OTFTs with BC configuration, because they are easier to fabricate,[14,26] but at the same time, BC OTFTs are more susceptible to contact effects than the TC configuration owing to the area of the contact.[5]

Since the unified model uses several material parameters, the electrode work function and the energy levels of the organic semiconductors are collected from the literature[21,24,29,32] and are summarized in Tables I and II. We consider the relative permittivity for organic materials to be around 3 ($\varepsilon_r\approx 3$) as reported for many materials in Refs. 24 and 30, the molecular density is $\rho_m=2\times 10^{21}$ cm^{-3} as reported in Ref. 24, and the contact length x_c is around ≈ 100 nm as deduced by noncontact scanning-probe techniques.[4] The mobility in organic materials is extracted by the characterization techniques detailed in Sec. III and by also using geometrical dimensions reported along with the experimental data in the literature.

TABLE I. Electrode work function of the metals referred to in this paper.

Electrode	Work function (eV)	Reference
Gold (Au)	4.9–5.47	24
Chromium (Cr)	4.5–4.8	32

A. Validation by the method using set of transistors with different channel lengths

The extraction method is detailed in Sec. III A. It is applied to output characteristics reported in Fig. 5a of Ref. 7 for BC pentacene OTFTs with Au contacts, channel width $W=270$ μm and channel lengths $L=2$, 10, and 20 μm. The extracted I_D-V_C curves are shown in Fig. 3(a) with symbols. For mobility $\mu=0.8$ cm^2 V^{-1} s^{-1}, the I_D-V_C curves converge in a straight line reproduced by the unified model. The calculations of the unified model prediction are detailed below.

The barrier height in these OTFTs is low, $\varphi_M=0.1$ eV, and the experimental currents are quite below the saturated current, $I_D^{\exp}\ll I_S=A^*T^2S\exp(-\varphi_M/kT)$, which results in small injection and redox voltages according to Eq. (4), $0\approx\exp[V_{\rm redox}/V_t(T)]\{\exp[V_{\rm injection}/\eta V_t(T)]-1\}$. The dominant phenomenon in this contact is therefore the drift and the voltage drop at the contact, V_C in Eq. (1), is $V_C\approx V_{\rm drift}$. As a result of these particular conditions, the electrical characteristics of these contacts are represented by linear I_D-$V_{\rm drift}$ curves, which can be deduced from Eq. (3), as explained below.

If the characteristic length x_p is a few times larger than the contact length x_c, we can expand in Taylor series, $x_p^{3/2}\{[1+(x_c/x_p)]^{3/2}-1\}=3/2x_cx_p^{1/2}+\cdots$, which reduces Eq. (3) to Ohm´s law

$$I_D\approx eSp(0)\mu\frac{V_{\rm drift}}{x_c}. \qquad (6)$$

According to our model, the charge density at the interface $p(0)$ equals to the value of the ion density ρ_{dp} and the prediction is calculated with Eqs. (2) and (6). After the calculation, $V_{\rm redox}\sim 2$ mV is verified to be much smaller than $V_{\rm drift}>100$ mV.

While above we studied an Ohmic contact in OTFTs and we demonstrated a reproduction of these contacts with the unified model, next, we will study non-Ohmic contacts in OTFTs, particularly the contact effects between chromium/gold electrodes and F8T2 with data from Ref. 5.

The I_D-V_C curves are extracted from the output characteristics given in Fig. 2 of Ref. 5 for transistors with different channel lengths. For mobility $\mu=6\times 10^{-3}$ cm^2 V^{-1} s^{-1}, the I_D-V_C curves converge, as shown in Fig. 3(b) with symbols. The unified model prediction (solid line) together with the injection component (dotted line) and the drift component (dashed line) are also depicted in Fig. 3(b).

TABLE II. Properties of the organic semiconductors referred to in this paper.

			Metal-organic interface		
Organic film	HOMO (eV)	Reference, year	Interface	Barrier height (eV)	Nernst potential
F8T2	5.5	29, 2004	Au/Cr-F8T2	0.68	$\Phi=0.22$ V
Pentacene	4.8–5.0	24, 2005	Au-Pentacene	0.1	$\Phi=0.39$ V
P3HT, P3HDT	≈ 5.2	21, 2005	Au-P3HDT	<0.4	$\Phi=0.30$ V

A nonlinear response of the contact can be observed. Street and Salleo[5] neglected the drift of the charge in the contact and proposed a diode-type relation as sufficient to reproduce the experimental measurements. As a result, they obtained a high value of the ideality factor $\eta=20$. This high value implies that all phenomena in the contact have to be combined. The parameters of the unified model with the prediction in Fig. 3(b), along with the calculations for each term, V_{drift}, V_{redox}, and $V_{\text{injection}}$, in Eq. (1), are detailed below.

The unified model can describe non-Ohmic contacts by using the terms related to charge drift and injection processes, Eqs. (3) and (4), respectively. A nonlinear relation between the current and voltage in the drift term occurs when the characteristic length x_p is much smaller than the contact length x_c and Eq. (3) reduces to the Mott–Gurney law,[22,23]

$$I_D \approx \frac{9}{8}\varepsilon\mu S\frac{(V_{\text{drift}})^2}{(x_c)^3}. \tag{7}$$

V_{drift} from Eq. (7) is shown by dashed line in Fig. 3(b), illustrating a parallel displacement (along the x-axis) with the experimental data (symbols) for the set of three chromium/gold-F8T2 OTFTs.[5] This parallel displacement suggests that the drift term calculated from Eq. (7) is appropriate for this set of data.

The fact that the drift term is characterized with Eq. (7) implies a short characteristic length x_p, which is obtained when $p(0)$ is large or redox voltages are small, according to Eqs. (2) and (3). The redox term is therefore small in these chromium/gold-F8T2 OTFTs, $V_{\text{redox}}\approx 0$. The small contribution of the redox reactions in these contacts makes the injection and the drift the main contribution to the voltage drop in the contact, $V_C \approx V_{\text{injection}}+V_{\text{drift}}$, according to Eq. (1). The injection term was defined in Eq. (4) and, for this particular case, $V_{\text{redox}}\approx 0$ is $I_D \approx A^*T^2S\exp(-\varphi_M/kT)\{\exp[V_{\text{injection}}/\eta V_t(T)]-1\}$, which explains the displacement between experimental data and the drift term in Fig. 3(b). A similar parallel displacement between a space-charge-limited current and experimental currents can be seen in Fig. 1 of Ref. 20. For that case, the authors also proposed the injection as a displacement between experimental data and the SCLC.

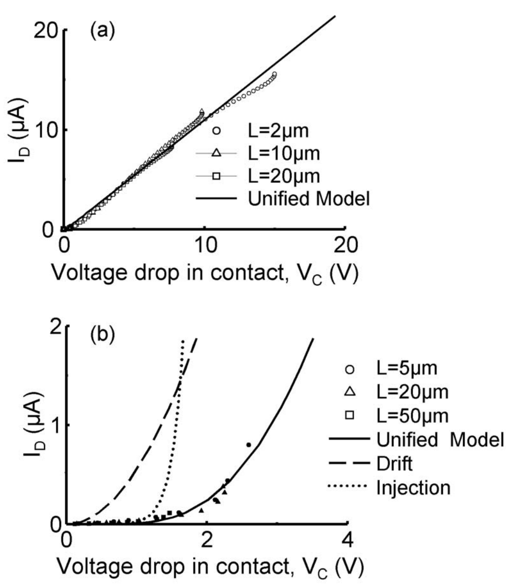

FIG. 3. I_D-V_C curves extracted with Eq. (5) and output characteristics, as described in Sec. III A. (a) Experimental I_D-V_C curves from Au-pentacene OTFTs data in Ref. 7 (symbols). The different channel lengths are $L=2$, 10, and 20 μm, $W=270$ μm, $x_C=100$ nm, $\mu=0.8$ cm^2 V^{-1} s^{-1}, the SiO_2 gate insulator thickness is 333 nm and $V_{GT}=V_{GS}-V_T=-20$ V. The barrier height between gold and pentacene is very low, making $V_{\text{redox}}\approx 0$ and $V_{\text{injection}}\approx 0$. The unified model prediction (solid line) from Eq. (1) is approximated by the drift term, Eq. (6), and reproduces well the experimental data. The thickness of the organic layer is $t_O=50$ nm and the Nernst potential is $\Phi=0.39$ V. (b) Experimental I_D-V_C curves from Au/Cr-F8T2 OTFTs data in Ref. 5 (symbols), the different channel lengths are $L=5$, 20, and 50 μm, $W=1000$ μm, $x_C=155$ nm, $\mu=0.006$ cm^2 V^{-1} s^{-1}, the SiO_2 gate insulator thickness is 200 nm and $V_{GT}=V_{GS}-V_T=-15$ V. The drift term (dashed line) is calculated from Eq. (7) to obtain a parallel displacement with the experimental data, the comparison between Eq. (7) and the general drift term in Eq. (3) suggests that the redox voltage is small. The injection term (dotted line) is then calculated from Eq. (4) and the unified model prediction (solid line) is the addition of the three terms. The barrier height is $\varphi_M=0.68$ eV, the thickness of the metal electrodes is $t_O=100$ nm, and the Nernst potential is $\Phi=0.22$ V.

B. Validation by the method using a set of characteristics of a single transistor biased at different gate voltages

Here, we apply the procedure from Sec. III B to extract the mobility and the contact voltage drop from the output characteristics of a single transistor at different gate voltages. We consider the output characteristics measured in poly(3-hexadecylthiophene) (P3HDT) transistors[25] (symbols in Fig. 4).

When the initial values for apparent mobility $\mu_A=5.5\times 10^{-5}$ cm^2/V s, $\alpha=0$, and threshold voltage $V_{AT}=-4.4$ V are used in the MOSFET model, a reasonable fitting can be obtained in the saturation region, as shown by solid lines in Fig. 4. However, the disagreement between this simplified model and the experimental curves in the linear regime is clear when neglecting the effect of contacts.

Taking the above initial values for μ_A and V_{AT}, the iterative procedure defined in Sec. III B is applied to determine the contact effects from the output curves of this OTFT. After

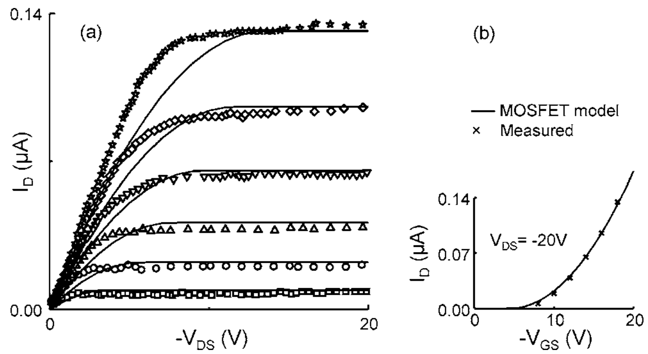

FIG. 4. (a) Output characteristics of an Au-P3HDT BC transistor at $V_{GS}=-8$, -10, -12, -14, -16, and -18 V from the bottom to the top. Symbols represent the experimental data. Lines represent the fitting with the simple MOSFET model without contact effects ($\mu_A=5.5\times10^{-5}$ cm^2/V s, $V_{AT}=-4.4$ V) and, evidently, the model deviates from the experimental data (Ref. 25). The details of the samples are: thermally grown SiO_2 ($\approx$200 nm), channel width W=1.5 cm (multifinger configuration), and channel length L=10 μm. The polymer film is 7 nm thick and was deposited by spin coating. (b) Transfer characteristics at $V_{DS}=-20$ V, used for an initial estimation of the mobility and the threshold voltage before iterations (the symbols represent experimental data and the line is the fitting with the simple MOSFET model).

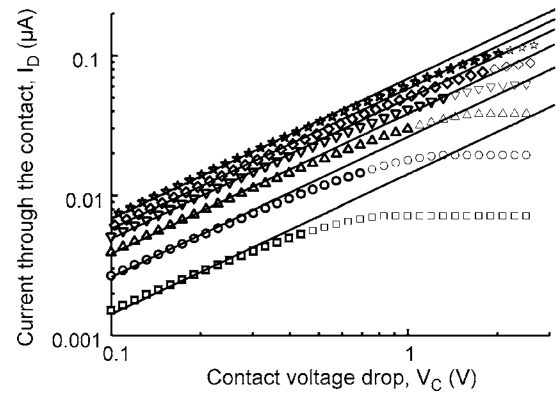

FIG. 6. Contact I_D-V_C characteristics of Au-P3HDT OTFT from Ref. 25. Symbols are extracted values from the output characteristics at different gate voltages after using the iteration method. Lines are calculated with the Ohmic prediction of the unified model, Eq. (6). From bottom to top, $V_{GS}=-8$, -10, -12, -14, -16, and -18, and $(V_{\mathrm{redox}}\text{-}\Phi)=-0.316$, -0.303, -0.295, -0.289, -0.285, and -0.282 V. The fabrication and electrical parameters are the same as in Figs. 4 and 5. Gray symbols illustrate the saturation regime of the TFT, in which the condition for use of Eq. (5) is violated.

the iteration procedure, the values of the intrinsic mobility μ and the contact voltage drop V_C are obtained. Substituting μ and V_C in Eq. (5), a very good fit at each gate voltage is obtained, as depicted in Fig. 5(a) in both saturation and linear regimes.

The values for $\alpha=V_C/V_{DS}=[R_C/(R_{ch}+R_C)]$ and μ are given in Figs. 5(b) and 5(c), respectively. Note that the mobility increases with the gate voltage as reported in the literature for different OTFTs.[11,16] The parameter α, which accounts for the ratio between the contact resistance and the channel resistance $\alpha=[R_C/(R_{ch}+R_C)]$, shows that the contact resistance decreases with the gate bias voltage. Decreases of the contact resistance, by charging the traps in the depleted region and increasing the concentration of free carriers, have been reported in the literature, by means of increasing the gate voltage[4] or by photogeneration.[25] The decrease in the contact resistance is explained by the increase in the induced charge carriers throughout the depleted region of the contact. The depleted region of the contact is rich in defects and they make the injected charge to get trapped, limiting the current flowing through the device. Thus, the greater the level of trapped charge by increasing the gate voltage or by illumination, the easier for the injected charge to travel across the depleted region without being trapped.

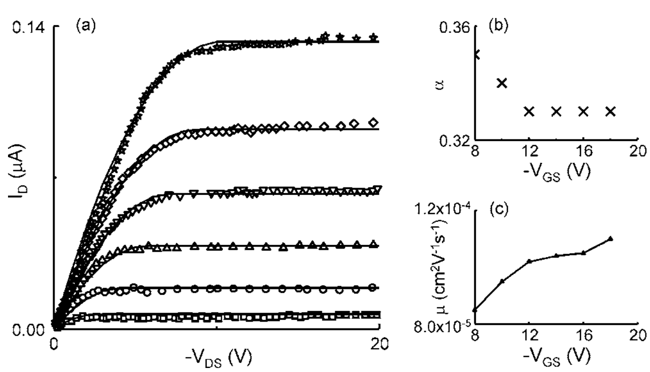

FIG. 5. (a) Experimental I_D-V_{DS} curves from Fig. 4 (symbols) and calculations of the I_D-V_{DS} curves (solid lines) taking into account the contact effects. The calculations are according to Eq. (5) with the parameter $\alpha=V_C/V_{DS}=[R_C/(R_{ch}+R_C)]$ represented in (b) and the mobility represented in (c) as a function of V_G.

Using the values of α and μ given in Figs. 5(b) and 5(c), the contact I_D-V_C characteristics of the Au-P3HDT OTFT are finally calculated. These I_D-V_C curves are shown in Fig. 6 with symbols and reveal that the contact in Au-P3HDT OTFTs is Ohmic. The prediction of the unified model is shown with solid lines in Fig. 6; it is calculated with Eq. (6) and reproduces the experimental data in the linear regime.

An Ohmic contact is expected owing to the small barrier height $\varphi_M<0.2$ eV between the gold electrodes and the P3HDT film. For this small barrier height, the contact voltage is mainly a result of the drift of charge, by the reasons discussed in the first part of the previous Sec. IV A. The redox and the injection terms are small compared to the drift term. That is why the prediction of the unified model, calculated from Eq. (6), reproduces the experimental data.

C. Limits of validity

A spurious divergence between the extracted I_D-V_C curves (symbols in Fig. 6) and the prediction of the unified model (lines in Fig. 6) occurs at high drain bias voltages. This disagreement is due to the operation of the transistor in the saturation regime, where the conditions for the approximation $V_C=\alpha V_{DS}$ used in the extraction procedure are violated. It is clear from Eq. (5) that the extraction of I_D-V_C curves is valid in the linear regime. The reason is that the gradual channel approximation is used in order to relate the drain current to the terminal voltages $[I_D=I_D(V_{GS},V_{DS})]$. Once this relation is defined, the contact effects are separated from the characteristics of the intrinsic transistor, $V_{DS}=V_C+V'_{DS}$.

This procedure, used in the linear regime, would still be valid in subthreshold or saturation regimes if proper expressions for the $I_D=I_D(V_{GS},V_{DS})$ curve are employed, and this will be investigated in our future work. In any of these three regimes, our model may be applied in order to reproduce the extracted I_D-V_C curves. No limits are imposed on our model outside the linear regime as the mobility and the charge density are the only variables in our model. Moreover, the mobility is not limited to a given voltage range because our model allows for the incorporation of different mobility models.

V. CONCLUSIONS

A unified model for the injection, interface dipoles, and transport of charge through contacts in organic transistors is presented. The combination of different physical phenomena in the unified model allows for the interpretation of a wide range of experimental data. Our unified model explains linear and nonlinear current-voltage curves at the contacts of organic transistors extracted by different characterization methods.

The ability of the unified model to fit various data for nonlinear contacts can help get deeper insights for the details and physical origin of the charge injection in organic devices, simultaneously providing a link to compact modeling.

ACKNOWLEDGMENTS

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